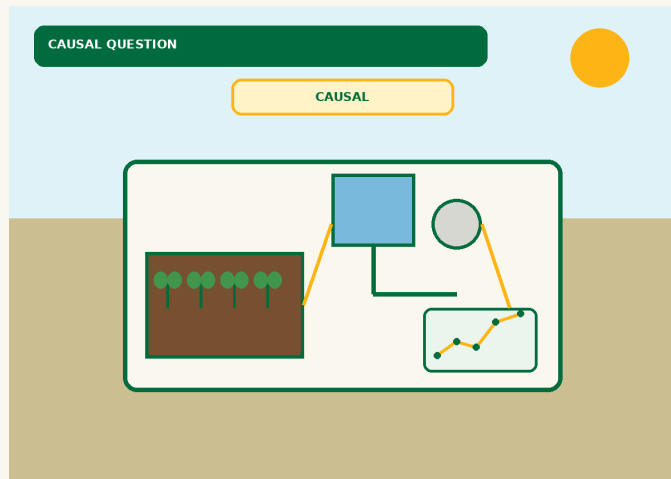




CAUSAL

CAUSAL QUESTION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

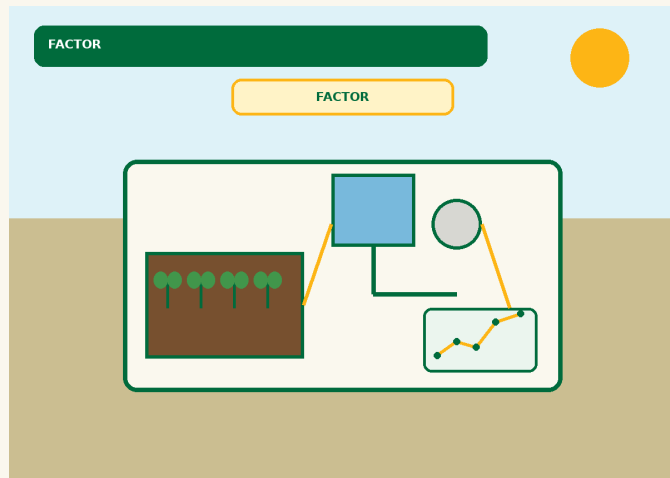
TECHNICAL CONCEPT

A controlled experiment is designed to estimate whether deliberately changing a factor causes a measurable response under specified conditions.

ANALYSIS / DESIGN TASK

Convert an observational question into a causal experimental question.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



FACTOR

FACTOR

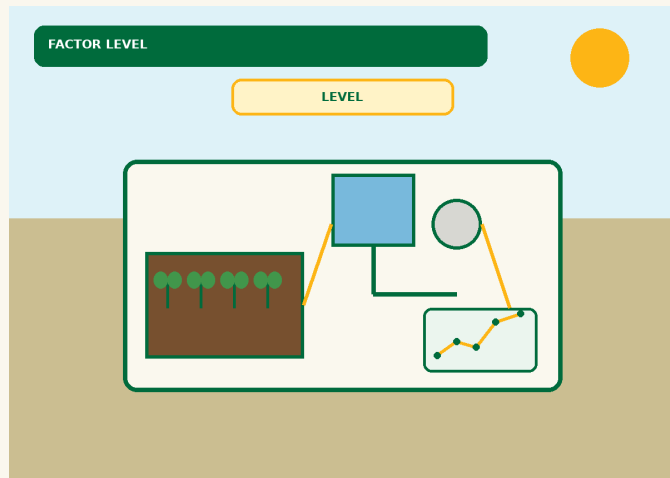
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A factor is an explanatory variable deliberately varied across two or more treatment levels.

ANALYSIS / DESIGN TASK

Define one experimental factor.



LEVEL

FACTOR LEVEL

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

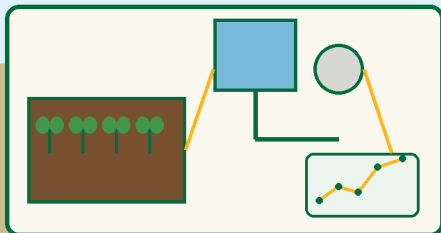
A factor level is a specific setting or category of the experimental factor.

ANALYSIS / DESIGN TASK

Specify three feasible levels.

FACTORIAL DESIGN

FACTORIAL



FACTORIAL

FACTORIAL DESIGN

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

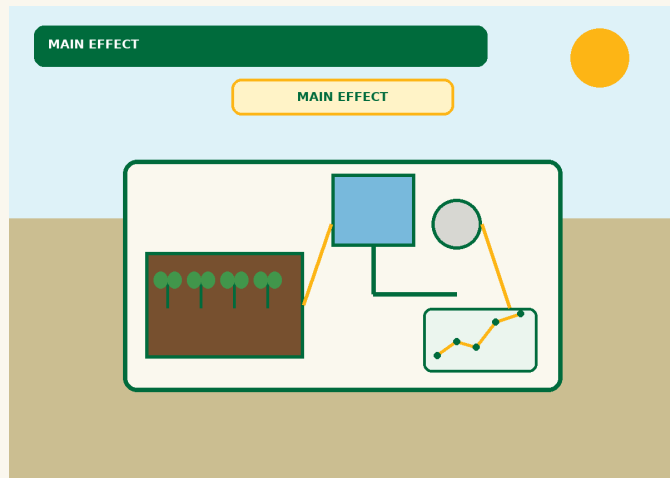
TECHNICAL CONCEPT

A factorial design evaluates two or more factors and their combinations within the same experiment.

ANALYSIS / DESIGN TASK

Construct a simple 2×2 treatment matrix.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



MAIN EFFECT

MAIN EFFECT

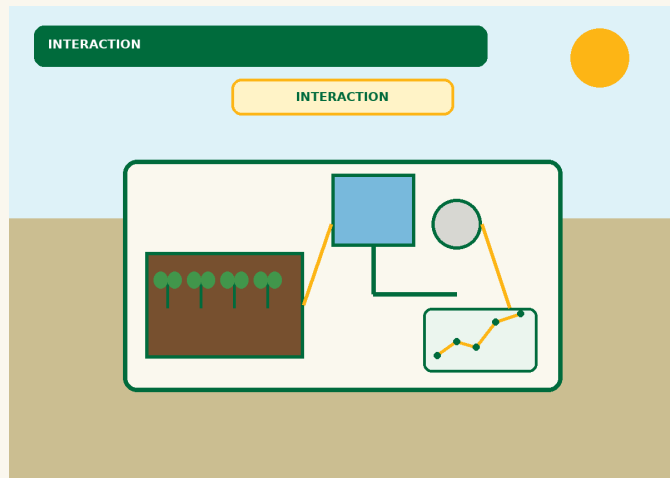
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A main effect is the average effect of changing one factor across levels of other factors in a factorial experiment.

ANALYSIS / DESIGN TASK

Interpret a teacher-provided main-effect example.



INTERACTION

INTERACTION

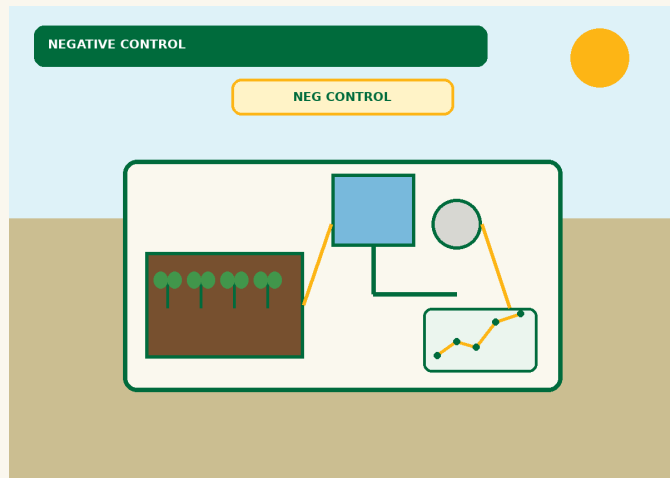
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

An interaction occurs when the effect of one factor changes depending on the level of another factor.

ANALYSIS / DESIGN TASK

Identify an interaction from a simple graph.



NEG CONTROL

NEGATIVE CONTROL

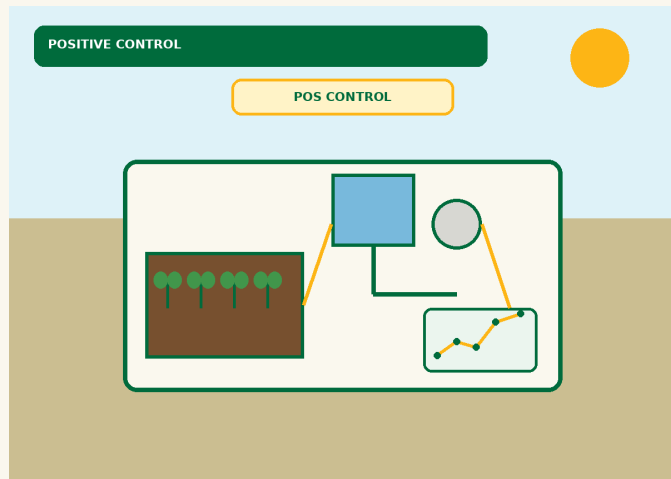
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A negative control is expected not to produce the tested response and helps reveal background effects or contamination.

ANALYSIS / DESIGN TASK

Explain its role in a safe plant experiment.



POS CONTROL

POSITIVE CONTROL

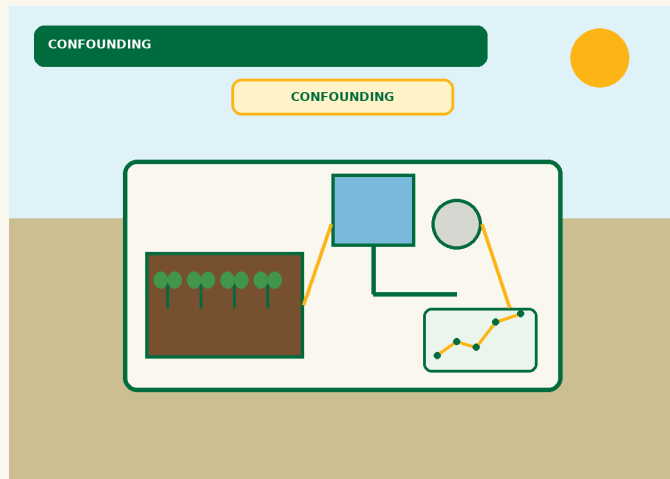
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A positive control is expected to produce a known response and helps verify that an experimental procedure can detect an effect.

ANALYSIS / DESIGN TASK

Choose a suitable positive-control concept from a scenario.



CONFOUNDING

CONFOUNDING

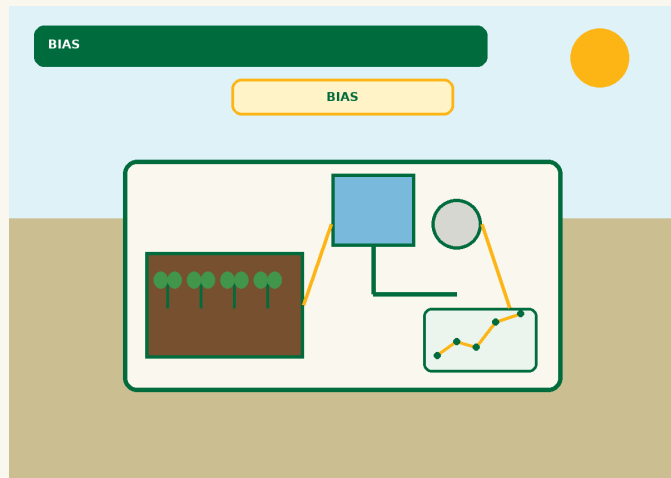
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Confounding occurs when the effect of a treatment cannot be separated from another systematically varying factor.

ANALYSIS / DESIGN TASK

Identify a confound in a flawed layout.



DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Bias is systematic error that shifts measurements, selection, treatment application, or interpretation away from the target quantity.

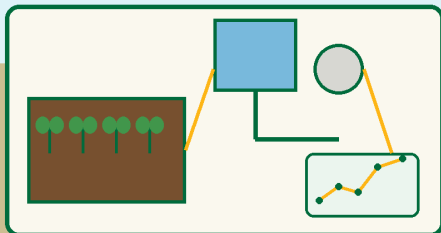
ANALYSIS / DESIGN TASK

Identify one source of bias and a prevention method.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

BLIND MEASUREMENT

BLINDING



BLINDING

BLIND MEASUREMENT

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

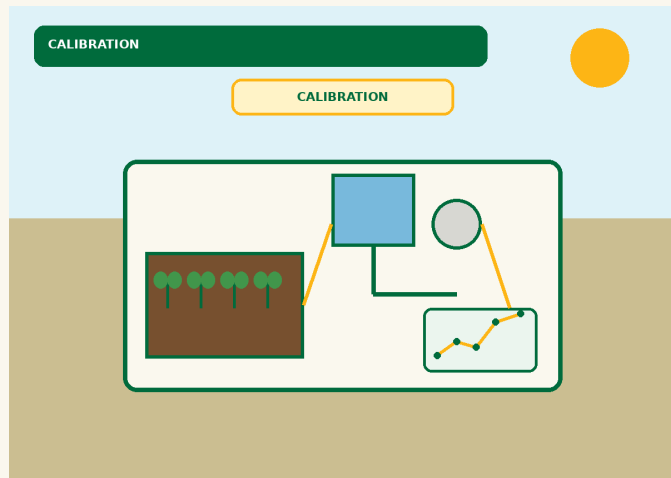
TECHNICAL CONCEPT

Blinding hides treatment identity from the measurer when feasible to reduce observer-related measurement bias.

ANALYSIS / DESIGN TASK

Design a coded measurement procedure.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



CALIBRATION

CALIBRATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Instrument calibration links measured output to accepted reference values before or during experimental data collection.

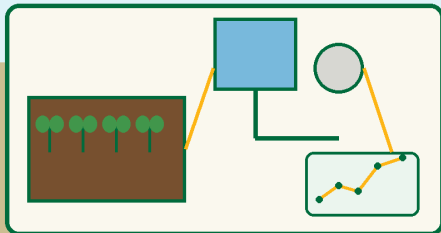
ANALYSIS / DESIGN TASK

Check an instrument against a reference.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

REPEATABILITY

REPEATABILITY



REPEATABILITY

REPEATABILITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

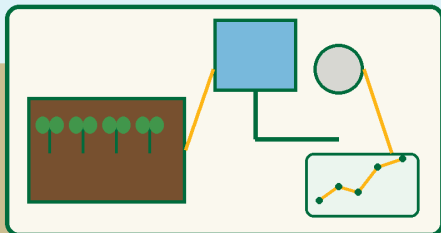
Repeatability describes agreement among repeated measurements made under the same conditions.

ANALYSIS / DESIGN TASK

Quantify spread among repeated readings.

REPRODUCIBILITY

REPRODUCIBILITY



REPRODUCIBILITY

REPRODUCIBILITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

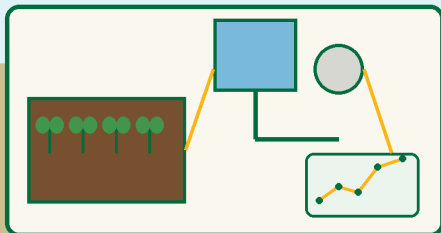
Reproducibility concerns whether a method or finding can be obtained again under clearly documented conditions or by another investigator.

ANALYSIS / DESIGN TASK

List documentation needed for reproduction.

QUALITY CONTROL

QC



QC

QUALITY CONTROL

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

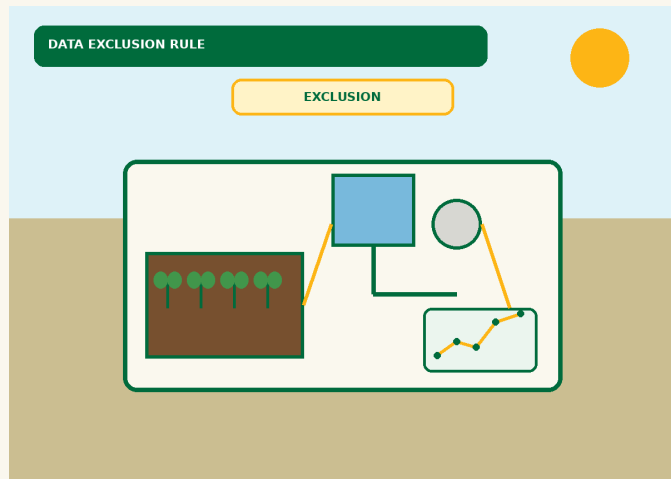
TECHNICAL CONCEPT

Quality control procedures detect and prevent errors during treatment application, measurement, recording, and data handling.

ANALYSIS / DESIGN TASK

Develop three QC checks.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



EXCLUSION

DATA EXCLUSION RULE

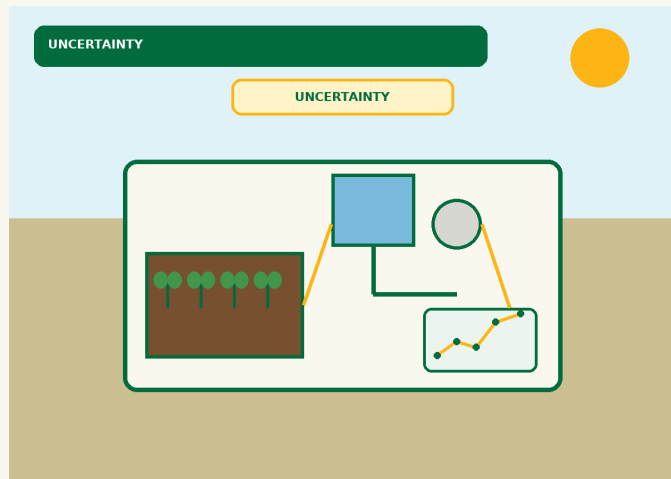
DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

A data-exclusion rule should be defined using objective criteria and not invented after seeing whether values support a preferred result.

ANALYSIS / DESIGN TASK

Draft a defensible pre-specified rule.



UNCERTAINTY

UNCERTAINTY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

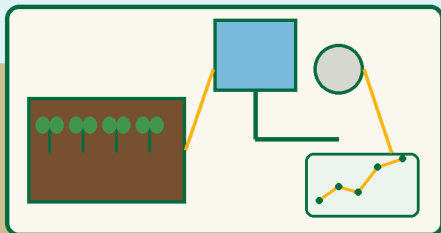
Measurement and experimental uncertainty describe limits on the precision and interpretation of estimated values and treatment effects.

ANALYSIS / DESIGN TASK

Identify two uncertainty sources.

SENSITIVITY ANALYSIS

SENSITIVITY



SENSITIVITY

SENSITIVITY ANALYSIS

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

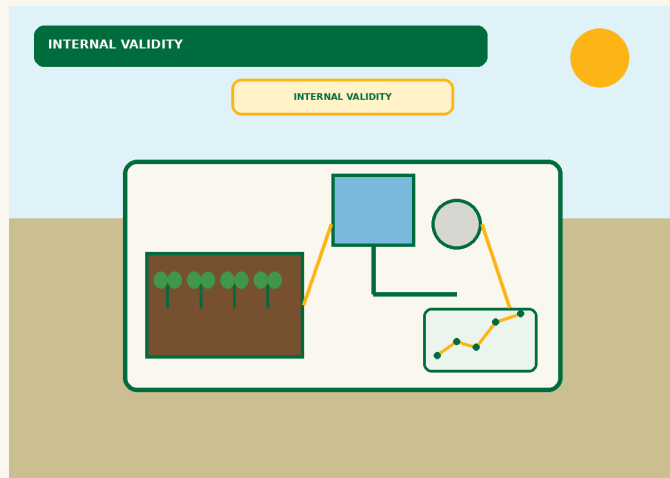
TECHNICAL CONCEPT

Sensitivity analysis examines how conclusions change when assumptions, parameter values, or analytical choices are varied.

ANALYSIS / DESIGN TASK

Test a conclusion under two assumptions.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE



INTERNAL VALIDITY

INTERNAL VALIDITY

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Internal validity concerns whether observed differences can credibly be attributed to the experimental treatment within the study.

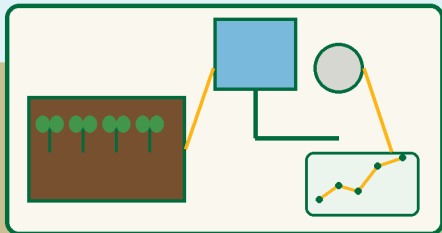
ANALYSIS / DESIGN TASK

Evaluate internal validity of a trial.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE

CONTROLLED EXPERIMENT EVALUATION

EVALUATE



EVALUATE

CONTROLLED EXPERIMENT EVALUATION

DESIGN • INSTRUMENT • MODEL • EVALUATE • OPTIMIZE

TECHNICAL CONCEPT

Evaluation integrates design, controls, bias, measurement quality, uncertainty, and causal interpretation.

ANALYSIS / DESIGN TASK

Audit a proposed experiment and recommend improvements.

DEFINE → MEASURE → MODEL → EVALUATE → DECIDE